Task 3 — Music Generation with AI

This project implements a small PyTorch LSTM that learns sequences of notes/chords extracted from MIDI files and samples a new sequence, which is exported as a MIDI file.

Important before running

- This is a training project, so **training data is not included**. Use MIDI files that you have permission to use.
- Training quality depends on the size and quality of the dataset. The included model is a starter baseline, not a professionally trained music model.
- Python 3.10 or 3.11 is recommended. PyTorch installation can vary by operating system and GPU; use the official PyTorch installation selector if the listed dependency does not install on your machine.

Setup (Windows PowerShell)

```
python -m venv .venv
.venv\Scripts\Activate.ps1
python -m pip install --upgrade pip
pip install -r requirements.txt
New-Item -ItemType Directory -Force data/midi
```

Copy your `.mid` or `.midi` files into `data/midi/` (subfolders are also searched). Then train:

```
python music_model.py train --epochs 20
```

Generate new notes:

```
python music_model.py generate --length 100
```

Output: `output/generated_music.mid`. Open it with a MIDI player or DAW. You can change the epoch count and generated length. The first training run may take some time.

Workflow

1. Parse MIDI files with `music21`.
2. Convert note/chord tokens into integer IDs.
3. Create sliding windows of 32 tokens.
4. Train an LSTM to predict the next token.
5. Sample predictions and export a generated MIDI file.

Troubleshooting

- `No MIDI files found`: add MIDI files to `data/midi/`.
- `Not enough notes`: add more MIDI material; at least 33 note/chord tokens are needed, and more data is strongly recommended.
- If PyTorch installation fails, install the appropriate build from the official PyTorch instructions for your OS/Python setup.